Fire Weather Index (FWI) Prediction System

Project Overview

This project is a machine learning-based web application designed to predict the risk of forest fires using environmental and fire weather index parameters. It allows users to input real-time data and get instant predictions along with visual insights such as charts and safety recommendations.

Objectives

1 To predict fire risk using a trained machine learning model.

2 To design a clean and user-friendly web interface.

3 To provide visual analysis using graphs.

4 To suggest precautions based on predicted risk.

System Architecture

The system consists of a frontend interface where users input data, a backend Flask application that processes the data, and a machine learning model that generates predictions. The results are displayed through a dashboard with graphs and insights.

Workflow

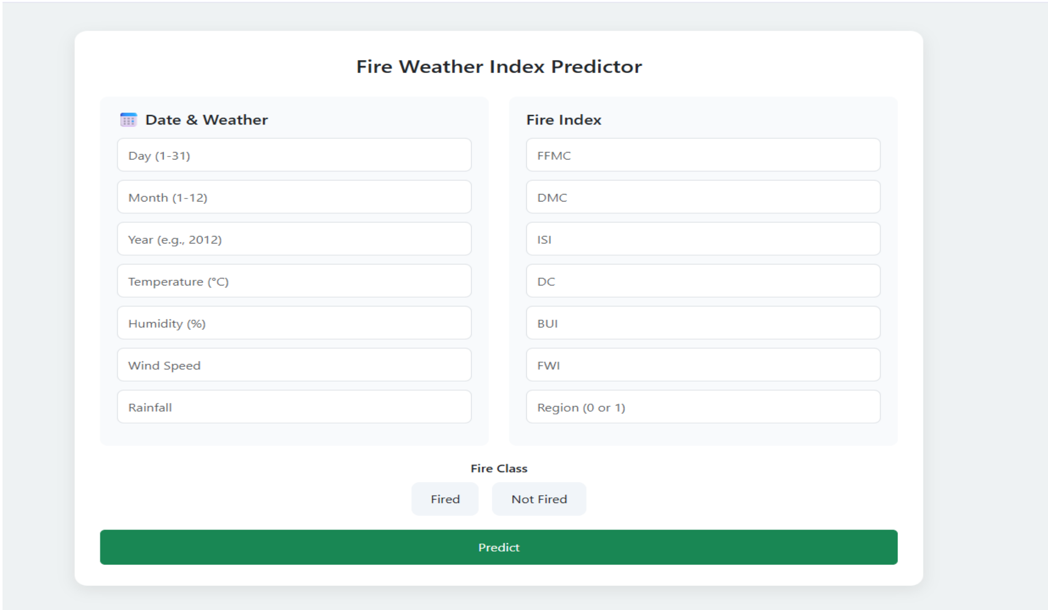
1 User enters environmental and fire index parameters.

2 Input data is validated and preprocessed.

3 Processed data is passed to the ML model.

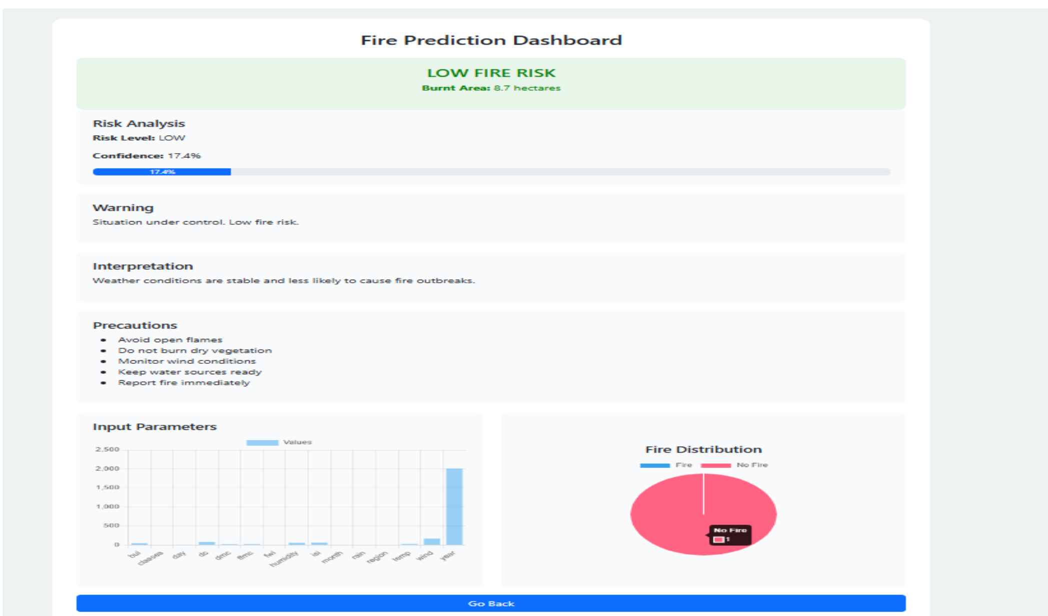
4 Prediction is generated and displayed.

Input Interface



The input interface is divided into sections such as Date & Weather and Fire Index. Users provide parameters like temperature, humidity, wind speed, rainfall, FFMC, DMC, ISI, DC, BUI, and FWI.

Output Dashboard



The output dashboard displays the predicted fire risk level, confidence score, warnings, interpretations, and precautions. It also includes graphical visualizations like bar charts and pie charts for better understanding.

End-to-End Pipeline

1. Input Stage: User provides required parameters.
2. Preprocessing: Data cleaning and scaling.
3. Model Processing: Prediction using ML model.
4. Output Stage: Results displayed with visuals.

Precautionary Measures

1. Avoid open flames.
2. Do not burn dry vegetation.
3. Monitor weather conditions.
4. Keep water sources ready.
5. Report fire immediately.

Conclusion

The system provides an efficient and user-friendly solution for predicting forest fire risks. It combines machine learning with a web interface to deliver accurate predictions and useful insights, helping in early prevention and awareness.